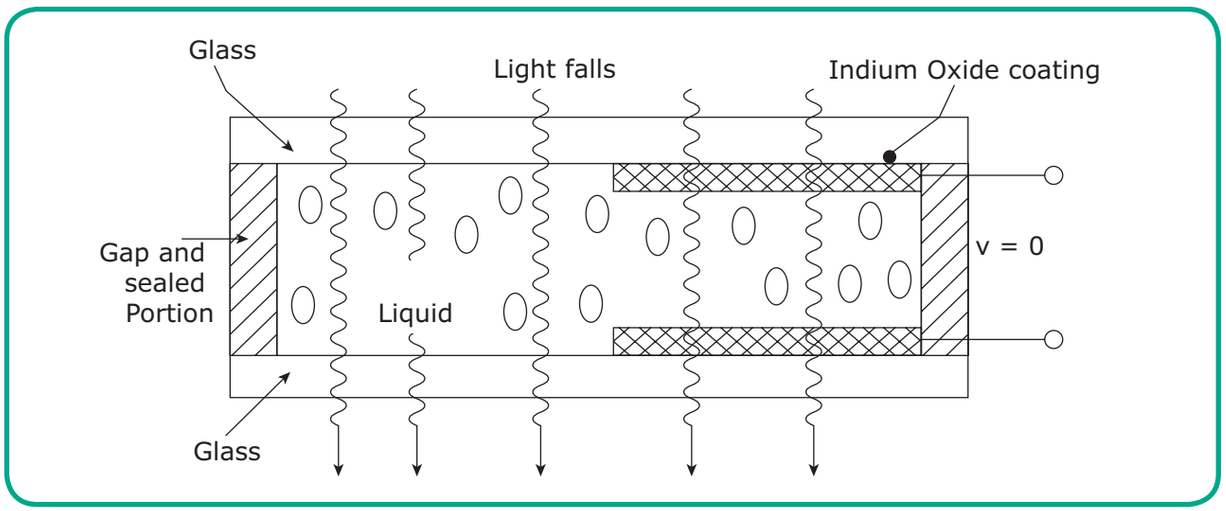


Figure 6.4(a) **LCD Construction**

inside with transparent electrodes separated by a liquid crystal layer of 5 to 50 μm thick. The oxide coating on the front sheet is etched to produce a single or multi-segment pattern of characters with each segment properly insulated from each other.

Working Principle: Figure 6.4(b) shows the LCD working principle**.** When AC voltage is not applied to the plates of the molecules of liquid crystal, which align themselves in perpendicular (or parallel) direction and appear transparent. When a voltage is applied across the conductive coatings, the arrangement of molecular pattern gets disturbed. As a result, refractive index of the medium of the crystal changes. Therefore, the incident light is reflected in different directions. Now, the liquid appears dark in the white background. The area not enclosed by conducting surfaces remains translucent.

Advantages of LCD

1. The operating voltage required is small (3 to 20 V/AC).
2. They are economical.
3. They have low power consumption.

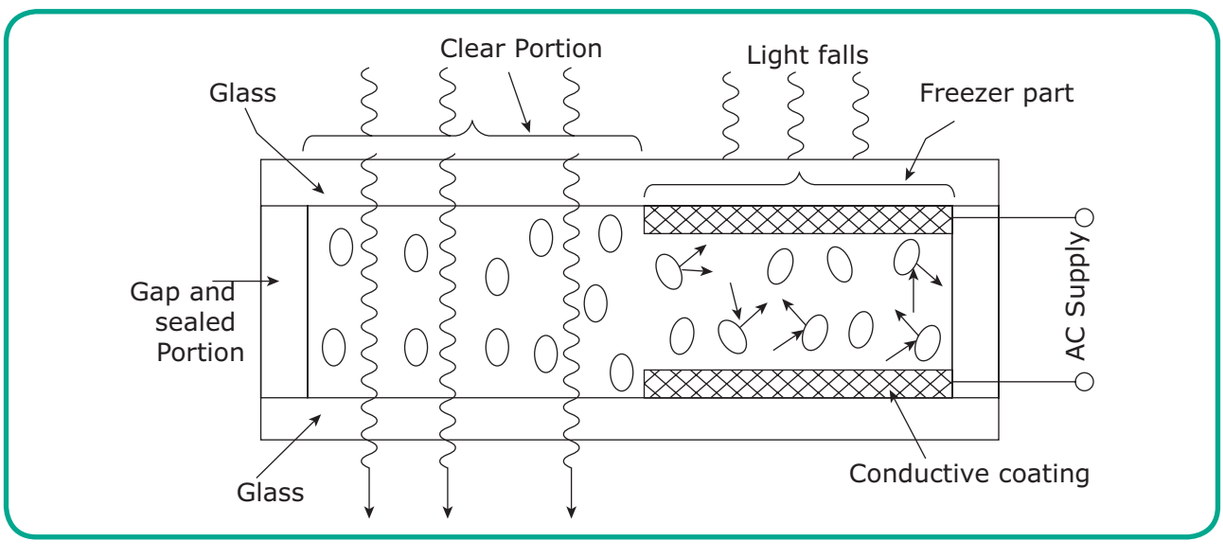


Figure 6.4(b) **LCD Working**

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